EDS 222 Midterm

Fall 2026

Name:

You have **75 minutes**. The exam is worth **100 points**:

- **Part 1:** five short questions, 10 points each. Aim to spend about 25 minutes on Part 1.
- **Part 2:** two analysis questions, 25 points each. Aim to spend about 20 minutes on each.

Write your answers in the space provided. If you run out of room, continue on the back of the page and say so. Show your work on calculations, because partial credit is available. Use the variable names given in each question when you write R code or model equations.

Question	Points	Score
1. Fill in the blanks	10	
2. Fix the error	10	
3. Model representations	10	
4. Interpreting inference	10	
5. Causal diagrams	10	
6. Analysis	25	
7. Analysis	25	
Total	**100**	

Reference

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) \qquad p = \frac{1}{1+e^{-\text{logit}(p)}}$$

$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \;/\; (p_{complex} - p_{simple})}{\text{RSS}_{complex} \;/\; (n - p_{complex})}$$

Backdoor criterion. A set of variables satisfies the backdoor criterion if adjusting for them (1) closes every backdoor path between x and y, and (2) doesn't include any variables caused by x.

Part 1: Short questions

1. Fill in the blanks (10 points)

A land trust measured soil carbon (`soil_c`, kg C/m²) in 25 meadows and how many years ago grazing ended (`years_ungrazed`). The data are in `meadows`.

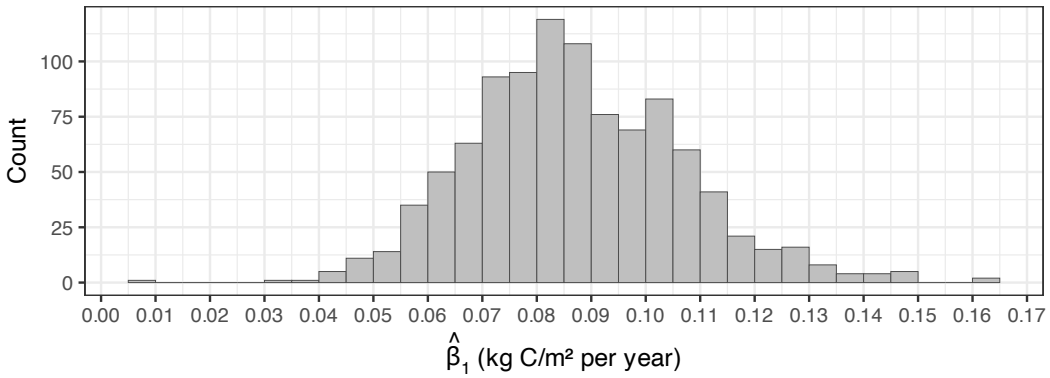
(a) (6 points) Fill in the blanks to calculate a 95% bootstrap confidence interval for β_1.

```
boot_dist <- replicate(1000, {
  boot_sample <- slice_sample(meadows, n = nrow(meadows), replace = __(1)__)
  coef(lm(soil_c ~ years_ungrazed, data = __(2)__))["years_ungrazed"]
})

ci <- __(3)__(boot_dist, c(0.025, 0.975))
```

(1) _____ (2) _____ (3) _____

(b) (4 points) The histogram shows `boot_dist`. Mark the 95% confidence interval on the histogram, then fill in the blanks (approximate values are fine).



We are 95% confident that β_1, the change in soil carbon per year without grazing, is between _____ and _____ kg C/m² per year.

2. Fix the error (10 points)

Ecologists counted bee visits (`visits`) in 45 plots in three habitats (`habitat`: **Grassland**, the reference level, **Shrubland**, and **Orchard**) and measured flower density (`flowers`). The data are in `plots`. Each part has **exactly one** error. Describe it and fix it. (5 points each.)

(a) Let the effect of flower density on visits differ among habitats.

```
visits_lm <- lm(visits ~ flowers + habitat, data = plots)
```

(b) An ANOVA comparing models with and without habitat has a p-value of 0.31.

> "Because p > 0.05, we accept H_0: habitat has no effect on bee visits."

3. Model representations (10 points)

Researchers recorded whether 400 coral colonies bleached (`bleached`: 1 = yes, 0 = no) and their heat stress (`dhw`, degree heating weeks). The data are in `corals`. Their fitted model is:

$$\begin{aligned} \text{bleached} &\sim \text{Bernoulli}(p) \\ \text{logit}(p) &= -4 + 0.6 \cdot \text{dhw} \end{aligned}$$

(a) (3 points) Fill in the blanks to fit this model.

```
glm(__(1)__, family = __(2)__(link = "logit"), data = corals)
```

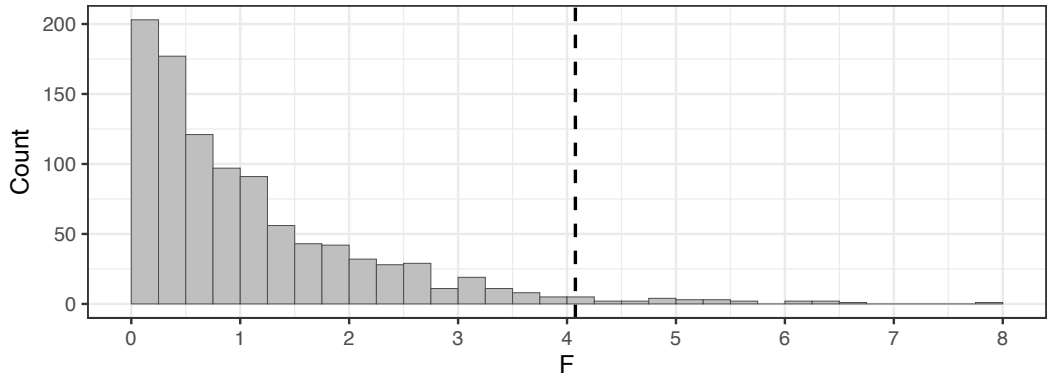
(1) _____ (2) _____

(b) (4 points) Calculate the probability that a colony bleaches when `dhw` = 10. Show your work.

(c) (3 points) At what `dhw` is a colony equally likely to bleach or not?

4. Interpreting inference (10 points)

Researchers measured soil respiration in 30 plots under three treatments (**Control**, **Compost**, **Biochar**). An F-test comparing `lm(respiration ~ treatment)` to an intercept-only model gave $F = 4.08$. The histogram shows F from 1,000 datasets with `treatment` shuffled. The dashed line is the observed F.



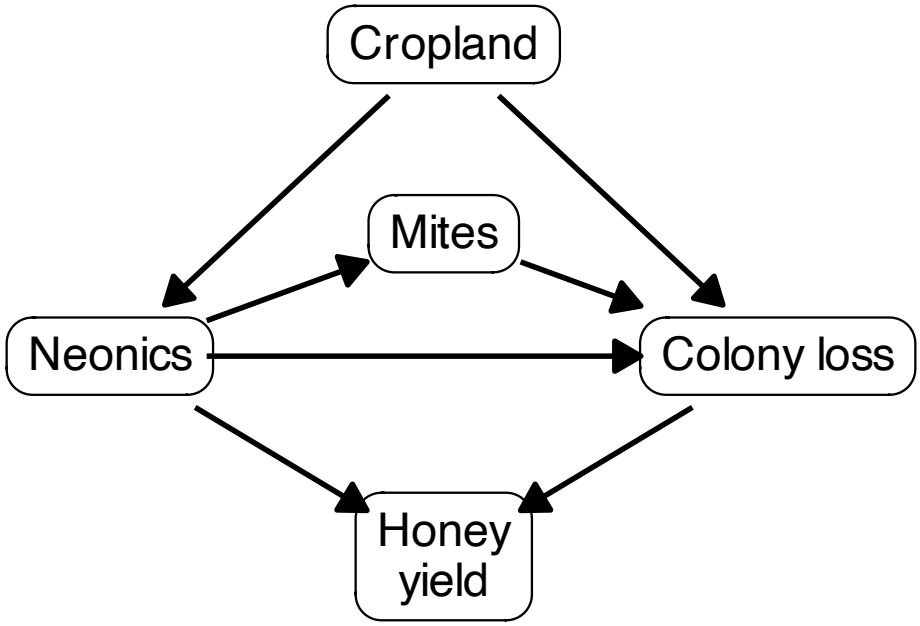
(a) (3 points) State H_0 in words.

(b) (3 points) Why does only the right tail count toward the p-value?

(c) (4 points) Is the p-value closer to 0.1 or 0.01? Do you reject H_0 at $\alpha = 0.05$?

5. Causal diagrams (10 points)

A state agriculture department asks: **how do neonicotinoid pesticides (Neonics) affect honey bee colony loss?** Here is their DAG for 120 apiaries. Neonicotinoids impair bees' foraging, so they lower each apiary's "Honey yield". Colony losses lower it too.



(a) (6 points) Is each variable a confounder, mediator, or collider for the effect of Neonics on Colony loss?

Cropland: _____ Mites: _____ Honey yield: _____

(b) (4 points) To estimate the **direct effect** of neonics (the part that *doesn't* work through mites), which variables should be in the model besides Neonics? Why?

Part 2: Analysis questions

6. Stream temperature and steelhead growth (25 points)

Warmer water speeds up a fish's metabolism. Up to a point, that means faster growth, but hatchery-raised fish may respond differently than wild fish. Biologists tagged 80 juvenile steelhead (`origin`: **Wild**, the reference level, or **Hatchery**) in reaches of a coastal river and measured each fish's **growth** (mm/day) over the summer and the mean water **temperature** (°C) in its reach. Hatchery fish were released throughout the river, so origin isn't related to temperature.

Riparian **shade** (% canopy cover) keeps reaches cool. Shade also boosts growth directly, because insects fall from overhanging trees into the water and fish eat them. Reaches with higher summer **flow** (m^3/s) are cooler, but flow doesn't affect growth any other way.

Question: how does temperature affect steelhead growth, and does the effect differ between wild and hatchery fish?

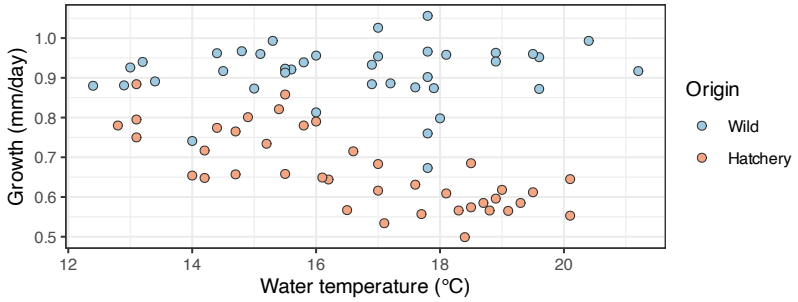
Variable	Description
`growth`	Growth rate over the summer (mm/day)
`temperature`	Mean summer water temperature in the fish's reach (°C)
`origin`	Wild (reference level) or Hatchery
`shade`	Riparian canopy cover over the reach (%)
`flow`	Mean summer flow (m^3/s)

(a) DAG (6 points). Draw a DAG with all five variables in the table.

(b) Choose variables (6 points). Which variables should you include in the model, and which should you leave out? For each, say why, using the terms confounder, collider, and mediator where they apply. Should the model include an interaction?

(c) Model (4 points). Write your model in statistical notation (use βs, not numbers), and define any dummy variables. Then write the formula you'd put in `lm()`.

Here's the data.



Three analysts each fit a different model.

```
fish_a <- lm(growth ~ temperature * origin, data = steelhead)
coef(summary(fish_a))
```

```
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)                0.8636    0.08755   9.864 3.01e-15
temperature                0.0029    0.00525   0.552 5.82e-01
originHatchery             0.3620    0.12547   2.885 5.08e-03
temperature:originHatchery -0.0365    0.00751  -4.860 6.17e-06
```

```
fish_b <- lm(growth ~ temperature * origin + shade, data = steelhead)
coef(summary(fish_b))
```

```
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)                0.11484   0.126285   0.909 3.66e-01
temperature                0.03860   0.006514   5.926 8.81e-08
originHatchery             0.32298   0.098099   3.292 1.52e-03
shade                      0.00389   0.000551   7.051 7.45e-10
temperature:originHatchery -0.03421   0.005873  -5.826 1.33e-07
```

```
fish_c <- lm(growth ~ temperature + origin + shade, data = steelhead)
coef(summary(fish_c))
```

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.35608   0.142835    2.49 1.48e-02
temperature    0.02358   0.007161    3.29 1.51e-03
originHatchery -0.24387   0.014933  -16.33 1.43e-26
shade          0.00406   0.000659    6.17 3.09e-08
```

```
anova(fish_c, fish_b)
```

```
Analysis of Variance Table

Model 1: growth ~ temperature + origin + shade
Model 2: growth ~ temperature * origin + shade
  Res.Df   RSS Df Sum of Sq    F  Pr(>F)
1     76 0.339
2     75 0.233  1     0.106 33.9 1.3e-07
```

(d) Inference (5 points).

i. (1 point) Which model matches your answer to (b)? Use it for the rest of this question.

ii. (2 points) The `anova()` output compares `fish_c` and `fish_b`. What are its H_0 and H_A? What do you conclude?

iii. (2 points) Compare the p-value from `anova()` to the p-value for `temperature:originHatchery` in `fish_b`. Why are they the same?

(e) Interpretation (4 points).

i. (2 points) Interpret the coefficient on `temperature` from your model in a sentence, with units. Which fish does it describe?

ii. (2 points) Calculate the effect of temperature on growth for hatchery fish. Show your work.

7. Hiking trails and invasive cheatgrass (25 points)

Cheatgrass is an invasive annual grass that fuels wildfires across the western US. Its seeds hitch rides on boots, pets, and horses, so trails may spread it. A national park surveyed 300 plots and recorded whether cheatgrass was present (`cheatgrass`: 1 = present, 0 = absent) and each plot's **distance** from the nearest trail (100s of m).

Trails follow valley floors, so plots near trails tend to be at lower **elevation** (km), and cheatgrass thrives in warm, low-elevation sites. Trails were built mostly through open **habitat** (Grassland, the reference level, Shrubland, or Woodland), and cheatgrass also grows best in open grassland. Trail use creates soil **disturbance** (% bare ground) near trails, which helps cheatgrass establish. Besides disturbance, trails also bring cheatgrass seeds into nearby plots directly.

Question: how does distance from a trail affect the probability that cheatgrass is present?

Variable	Description
`cheatgrass`	1 if cheatgrass is present in the plot, 0 if absent
`distance`	Distance to the nearest trail (100s of m)
`elevation`	Elevation (km)
`habitat`	Grassland (reference level), Shrubland, or Woodland
`disturbance`	Bare ground in the plot (%)

(a) DAG (6 points). Draw a DAG with all five variables.

(b) Choose variables (5 points). We want the **total effect** of distance on cheatgrass. Which variables should you include, and which should you leave out? Say why, using the terms confounder, collider, and mediator where they apply.

(c) Model (4 points). Write your model in statistical notation (use βs, not numbers). Include the distribution of the response, and define any dummy variables. Then write the formula you'd put in `glm()`, and its `family` argument.

Three analysts each fit a different model.

```
cheat_a <- glm(
  cheatgrass ~ distance,
  family = binomial(link = "logit"),
  data = plots
)
coef(summary(cheat_a))
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)     1.80      0.306    5.86 4.68e-09
distance       -1.61      0.243   -6.64 3.14e-11
```

```
cheat_b <- glm(
  cheatgrass ~ distance + elevation + habitat,
  family = binomial(link = "logit"),
  data = plots
)
coef(summary(cheat_b))
```

```
                Estimate Std. Error z value Pr(>|z|)
(Intercept)        3.134      0.635    4.93 8.08e-07
distance          -0.870      0.351   -2.48 1.33e-02
elevation         -1.003      0.400   -2.51 1.22e-02
habitatShrubland  -0.382      0.347   -1.10 2.71e-01
habitatWoodland   -0.988      0.421   -2.35 1.89e-02
```

```
cheat_c <- glm(
  cheatgrass ~ distance + elevation + habitat + disturbance,
  family = binomial(link = "logit"),
  data = plots
)
coef(summary(cheat_c))
```

```
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      3.00237     1.0364    2.90  0.00377
distance        -0.83597     0.4120   -2.03  0.04246
elevation       -1.00010     0.4009   -2.49  0.01261
habitatShrubland -0.38684     0.3484   -1.11  0.26691
habitatWoodland  -0.99872     0.4265   -2.34  0.01918
disturbance       0.00433     0.0271    0.16  0.87299
```

(d) Inference (5 points).

i. (1 point) Which model matches your answer to (b)? Use it for the rest of this question.

ii. (2 points) State H_0 and H_A for the effect of distance on cheatgrass, in words and in terms of a β.

iii. (2 points) Interpret the p-value for `distance` in a sentence. Do you reject H_0?

(e) Interpretation (5 points).

i. (2 points) Interpret the coefficient on `distance` from your model in a sentence, with units.

ii. (3 points) Calculate the probability of cheatgrass in a **shrubland** plot 100 m from a trail at 1.5 km elevation. Show your work.